

RUDRA SHARMA

Associate AI/ML Engineer | Data Engineer

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SUMMARY

AI/ML and Data Engineer building production RAG, agentic AI, and data platforms (Python, FastAPI, Spark, Databricks, PostgreSQL, Docker). Shipped a 15–20x faster data engine and a 50% marketing-overhead cut at BytePX, a 95% latency cut for a RAG chatbot at UltraTech, plus a hybrid RAG platform and the AI-OS agent framework.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript

ML/AI: Scikit-learn, TensorFlow, PyTorch, XGBoost, SHAP, LLMs, RAG, Agentic AI, MCP, LangChain, Qdrant, NLP

Data Engineering: Apache Spark, Databricks, Delta Lake, Pandas, ETL/ELT Pipelines, Data Validation

Backend & Cloud: FastAPI, Flask, REST APIs, PostgreSQL, Docker, Git, GitHub, Microsoft Azure, IBM watsonx

EXPERIENCE

Associate ML Engineer – BytePX

March 2026 – Present

- **DataPurge Studio:** Architected an async, multi-tenant SaaS data engine (FastAPI, PostgreSQL, Docker) that processes complex multi-sheet workbooks in under 2 minutes at 99.9% accuracy, a 15–20x speedup.
- **Data Tool:** Built a multi-platform integration tool that ingests from 5 sources (Salesforce, Amazon S3, Google Sheets, Dynamics 365, cloud storage), applies cleansing, schema-mapping, and validation, and exports to comparable platforms; added an AI agent that runs these tasks from natural-language prompts.
- **Client ML delivery:** Cleaned client datasets and trained a Random Forest model with SHAP explanations that reduced marketing overhead by 50%; deployed proofs of concept on IBM watsonx.
- **DataPurge performance:** Cut fuzzy-matching time 10x with an O(N) lookahead sniffer and RapidFuzz (C++), streaming data through io.BytesIO buffers and Starlette thread pools.
- **Databricks/Spark pipelines:** Automated multi-format document processing and contract tracking, including a resume converter that cut processing time by 90%.
- **HR Hours Reconciliation:** Automated ERP-to-employee hours verification end to end (ERP fetch, bot-based employee confirmation, reconciliation), eliminating 8–10 hours of manual work per month across 20+ employees.
- Mentored 2 junior engineers on Python, Spark DataFrames, and Databricks across 3 internal and client projects.

AI & ML Intern – UltraTech Cement Ltd. [Birla White]

June 2025 – July 2025

- Built “Birbal 2.0”, a multilingual agentic RAG chatbot (FastAPI) for regional technical teams, cutting query response latency by 95% by replacing blocking SQL/API calls with asynchronous retrieval.
- Added Hindi-English support and contextual state tracking, improving query efficiency by 60%.

Generative AI Intern – SmartBridge (concurrent)

June 2025 – July 2025

- Built GenAI prototypes (multilingual chatbots, predictive models) using FastAPI and Google Gemini APIs.

SELECTED PROJECTS

VoxContextEngine – Production Hybrid RAG Platform

VOX

- Engineered a production-grade hybrid RAG platform (FastAPI, Docker) fusing dense search (Qdrant, all-MiniLM-L6-v2, HNSW) with BM25 to ground every LLM answer in source documents and curb hallucination.
- Built an automated Python evaluation suite that benchmarks ingestion and retrieval precision, validating hallucination resistance with a 100% safety-run completeness score.

AI-OS – Versioned Operating Layer for AI Coding Agents

AI-OS

- Designed a versioned operating layer for AI coding agents (Claude, Gemini, Codex): adapter files, project playbooks, and a self-improvement loop that promotes only quality-checked lessons into one canonical rule hierarchy.
- Cut agent latency up to 5x and token cost 90% with pinned-prefix prompt caching and multi-agent MCP orchestration; a self-healing test loop auto-remediates failures before escalation.
- Added a pre-commit secret-interception hook and 14-point health check for zero-incident credential handling.

EDUCATION & CERTIFICATIONS

UPES – B.Tech in Computer Science (AIML), CGPA: 8/10

Aug 2022 – May 2026

Certification: Microsoft Certified: Azure AI Engineer Associate (Credential: 957CF25F904A0B46)